



Semester 1 Examinations 2013 / 2014

Exam Code(s)	1SD1, 1MF1, 1SD3
Exam(s)	Higher Diploma (Software Design & Development) M.Sc. (Software Design & Development) Higher Diploma (Software Design & Development) – Industry Stream
Module Code(s)	CT516 CT861 CT538 CT542
Module(s)	Computer Architecture and Operating Systems
Paper No.	I
External Examiner(s)	Professor Liam Maguire
Internal Examiner(s)	Professor Gerard Lyons *Dr. Petronel Bigioi *Dr. Enda Howley *Dr. Matthias Nickles *Dr. Michael Schukat

Instructions: CT516 and CT861: Answer 2 questions from Section A and one question from section C.

CT538 and CT542: Answer 2 questions from Section A and one question from section B.

<i>Duration</i>	2 hrs
No. of Pages	5
Discipline(s)	Information Technology
Requirements	None

Section A

Question 1

- a) Distinguish between *distributed*, *embedded*, *desktop* and *mobile computers*. Where can these types of computers be found in the area of *everyday activities* or *transport*? Provide one example per computer type.

8 MARKS

- b) How does interrupt processing in a CPU work? In your answer refer also to the Instruction Register (IR) and the Program Counter (PC).

9 MARKS

- c) Using a detailed example show the benefits of a CPU with *multiple* general purpose registers over a CPU with only *one* general purpose register (e.g. the Accumulator).

8 MARKS

Question 2

- a) What is meant by the term *locality of reference*? How does it relate to the concept of *caching* and to *cache design*?

11 MARKS

- b) What is the difference between the *sign-magnitude* and the *two's-complement* representation of a negative integer? Explain which representation in your view is the better one.

6 MARKS

- c) Pick two positive (e.g. > 0) 4-bit long integers of your choice and show how binary addition works. Discuss how a numeric overflow in the sum of the 2 values can be detected.

8 MARKS

Question 3

- a) What is a *combinatorial circuit*? Use a comprehensive example to illustrate your answer.

11 MARKS

- b) Assume you design a CPU with 20-bit instructions where each instruction has 2 operand references. Draw up a *suitable* instruction format and calculate how many instructions and operand references you can distinguish.

7 MARKS

- c) Using an example outline the main two factors that determine the access time of a hard disk.

7 MARKS

PTO

Question 4

- a) A problem which can occur in the context of memory management is *external fragmentation*. Explain what external fragmentation is and how this problem can be minimized.

6 MARKS

- b) Describe three placement algorithms for *dynamic partitioning*. What are the advantages and disadvantages of each of these algorithms?

12 MARKS

- c) What does it mean to say that a process is “*suspended*”? Use examples to describe situations in which it is useful or required to suspend a process temporarily.

7 MARKS

Section B

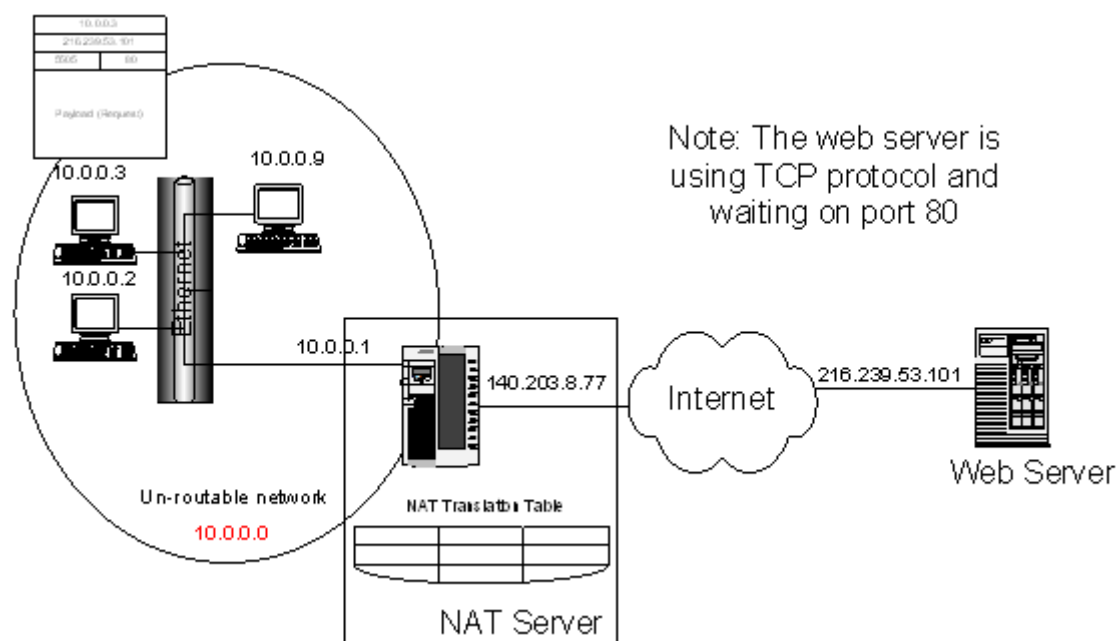
Question 5

- a) Describe what happens following a collision in Ethernet (802.3). Some LAN technologies e.g. Token Ring (802.5) provide a mechanism for priority access, can you outline a similar facility available in Ethernet (802.3)?
7 MARKS
- b) Describe at least two modulation techniques?
8 MARKS
- c) Explain briefly the difference between a connection-oriented and connectionless network. In this context, which type of network is the Internet itself (using the IP protocol).
10 MARKS

Question 6

- a) Consider the network topology in the figure below.
 - Describe the operation of NAT assuming that the host 10.0.0.3 in the un-routable network 10.0.0.0 makes a web request to the server 216.239.53.101 (web server is waiting on port 80)
 - Show how the request TCP/IP packet is modified on its way from host 10.0.0.3 to the web server.
 - Show how the reply packet from the server 216.239.53.101 to host 10.0.0.3 is modified by NAT.

25 MARKS



Section C

Question 7

- a) Describe the key properties of three different algorithms for *disk scheduling* (for conventional data). Then, assume a disk has 200 tracks (track numbers 0 to 199) and that the following sequence of track accesses should be completed: 20, 190, 50, 60, 10, 40. Calculate (for each of the three algorithms) how many tracks need to be traversed in order to complete the entire sequence (assume the read/write head is initially at track 0).
- 13 MARKS
- b) Describe what *polling* is (in the context of I/O). Which alternative approach is typically preferable, and why?
- 6 MARKS
- c) Describe the role of an *I/O scheduler*.
- 6 MARKS

Question 8

- a) What is an *Access Control List* (ACL)? To illustrate your English explanation, sketch an *Active Matrix* for three users (uA, uB, uC) and four files (f1, f2, f3, f4). uA owns f1 and has read and write access to f2 and f3. uB has read access to f3 and f4. uC has write access to all files. No other rights and ownerships exist. Mark all Access Control Lists in this matrix.
- 10 MARKS
- b) Explain what *chained allocation* is. Also describe a potential problem caused by chained allocation and explain how this problem can be addressed.
- 9 MARKS
- c) Name and describe two major objectives of I/O device management.
- 6 MARKS